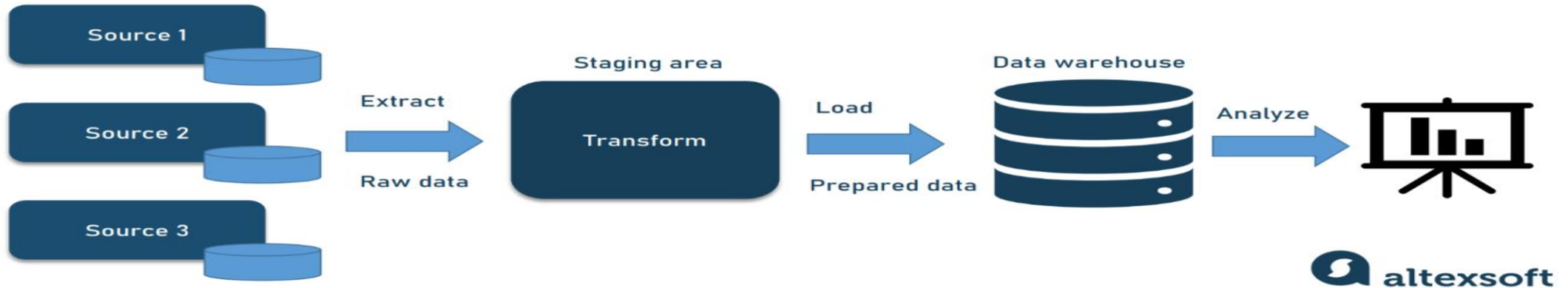




- Title:** Data Science Capstone Project
- Subtitle:** SpaceX Launch Analysis
- Name:** Tarun

ETL PIPELINE



- Goal: Predict whether SpaceX rocket will land successfully
- Why important:
- Cost saving
- Reusability

Data Cleansing Process



- Removed missing values
- Converted data types
- Created new columns
- Cleaned inconsistent data

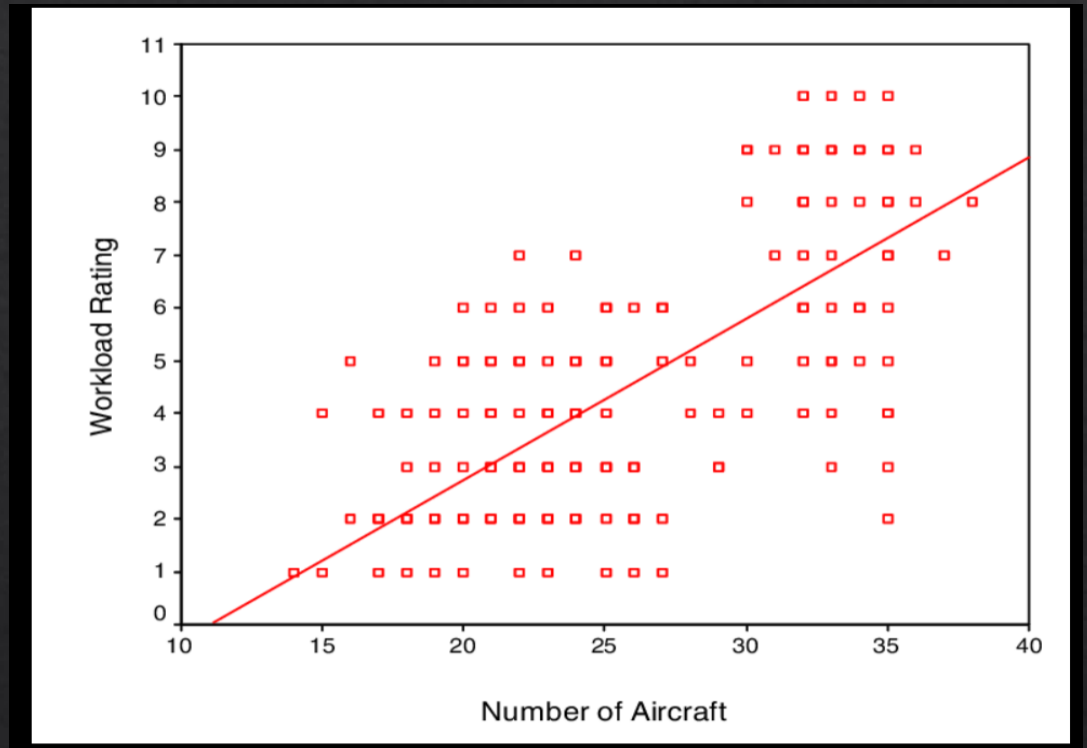
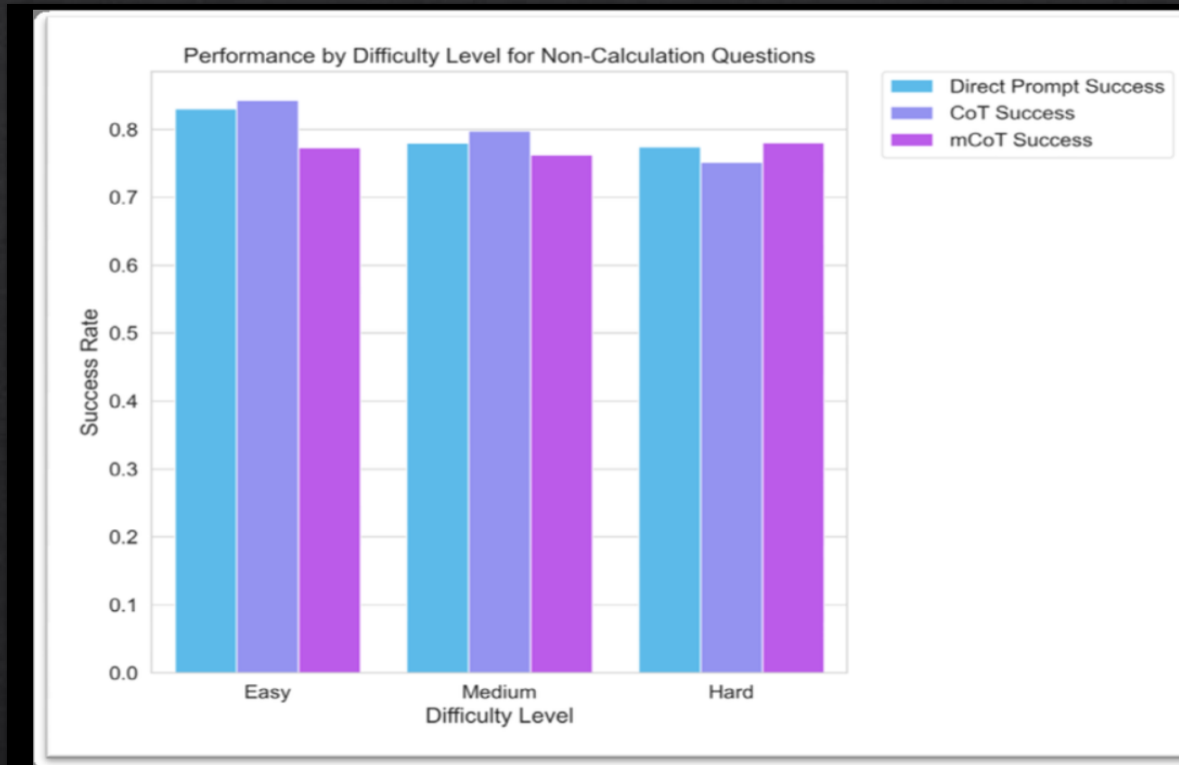
- Handled missing values using `dropna()` and `fillna()`
- Removed duplicate records using `drop_duplicates()`
- Converted data types using `astype()`
- Renamed columns using `rename()`
- Filtered relevant data using conditions
- Created new columns using feature engineering
- Used `groupby()` for data aggregation

Encrypting Database SQL Server



- ◆ #Total launches
`SELECT COUNT(*) FROM launches;`
- #Success launches
`SELECT COUNT(*)
FROM launches
WHERE landing_outcome='Success';`
- # Launch sites
`SELECT DISTINCT launch_site FROM launches;`
- #Most used rocket
`SELECT booster_version, COUNT(*)
FROM launches
GROUP BY booster_version;`

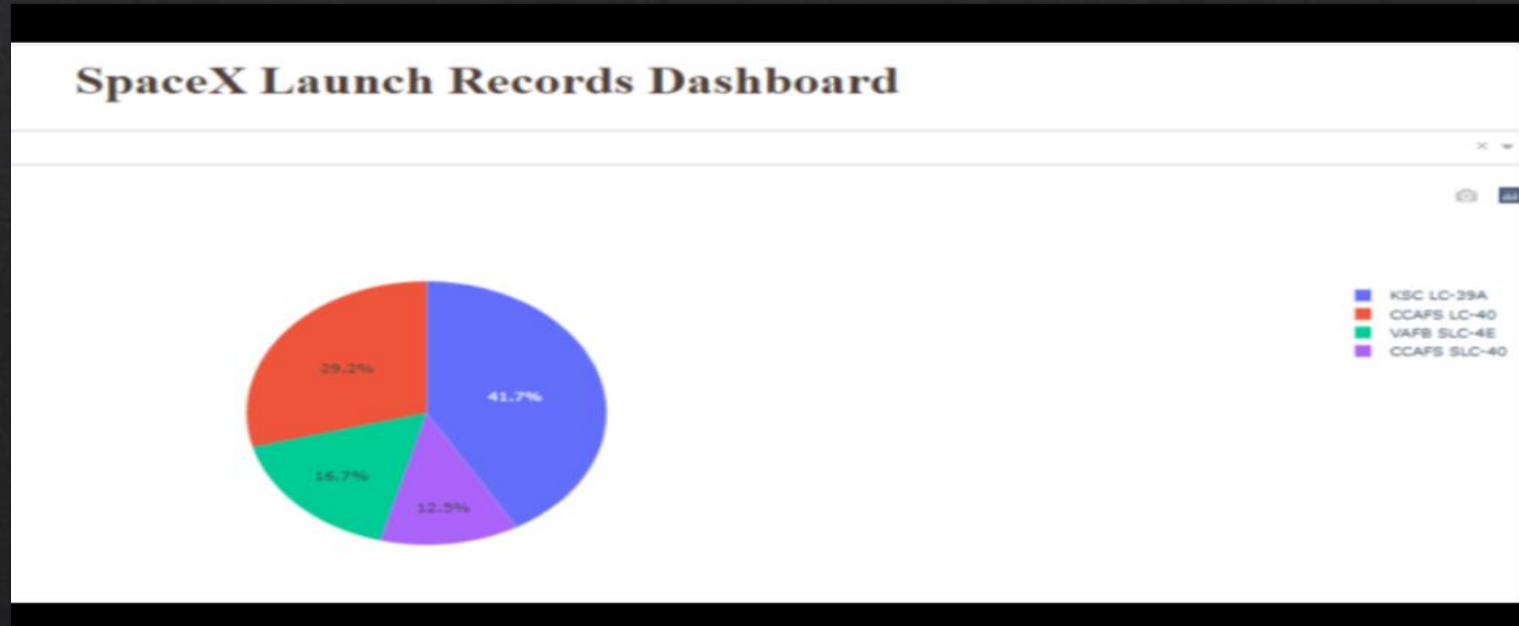
DATA VISUALIZATION



- Success rate increases with experience
- Some launch sites perform better

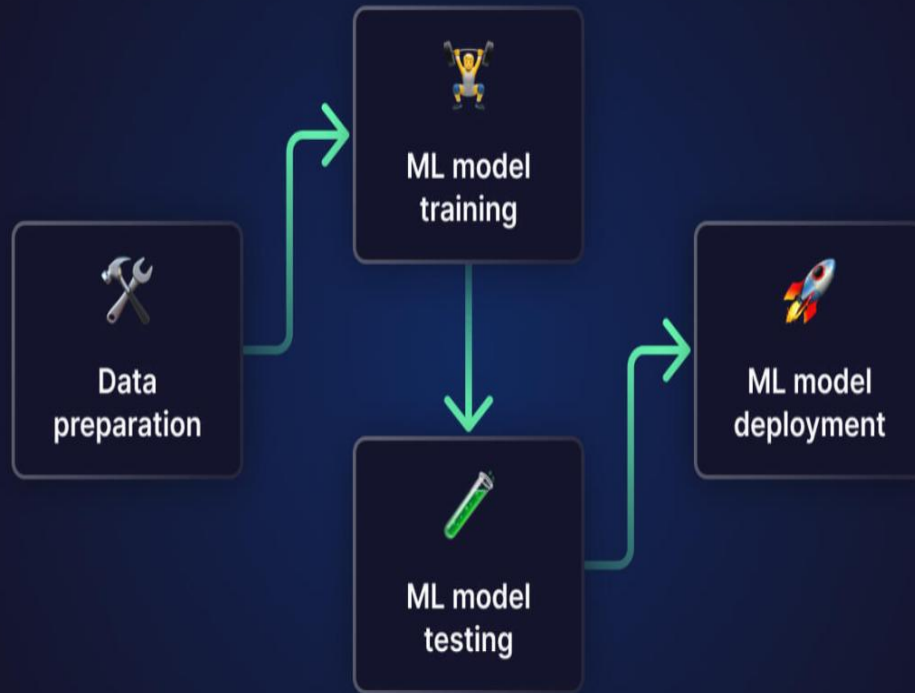
- **Bar Chart**
- X: Launch Site
- Y: Success Rate
- **Pie Chart**
- Success vs Failure
- **Scatter Plot**
- Flight Number vs Payload

SpaceX Launch Record Dashboards



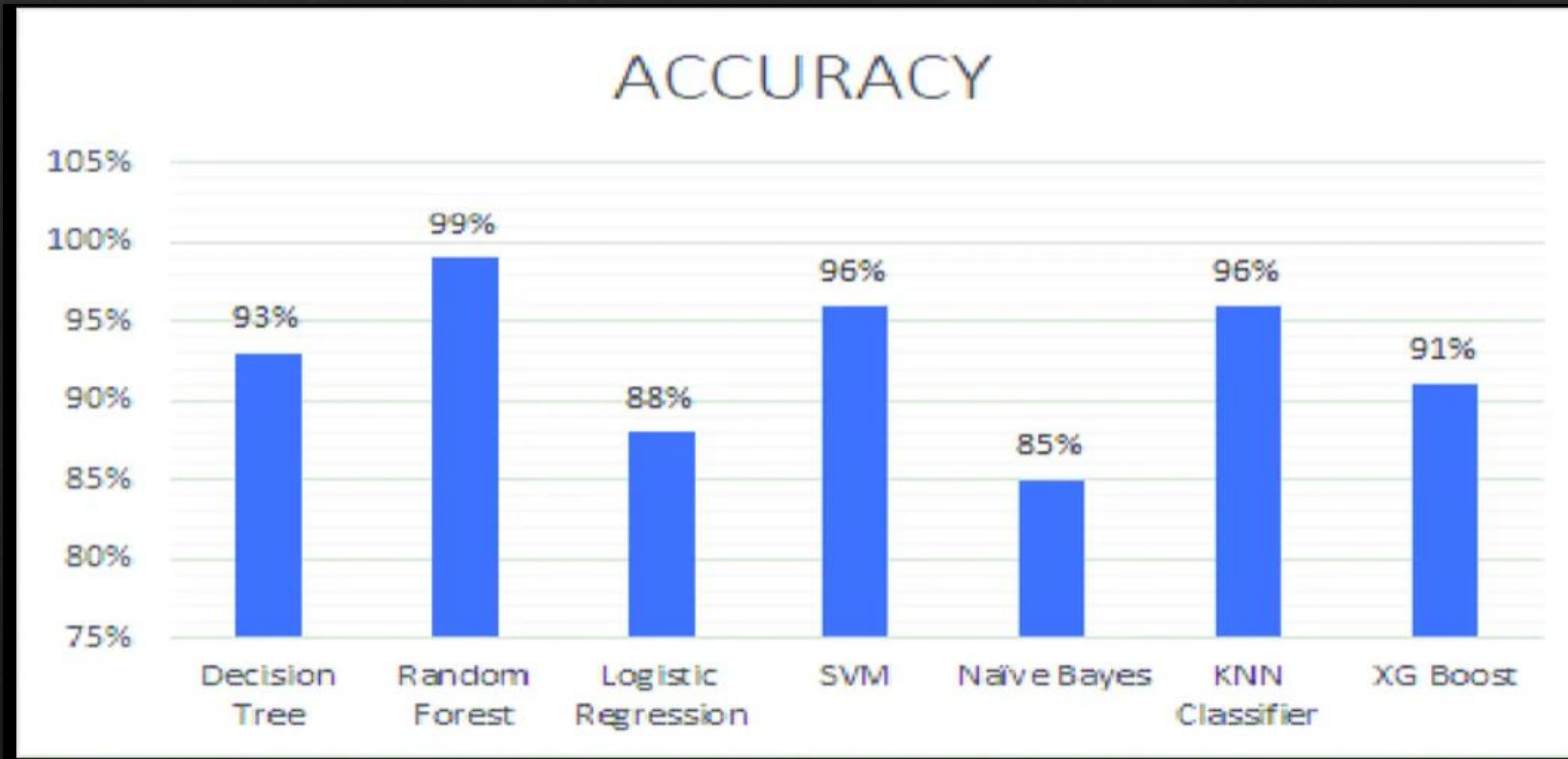
- Built using Plotly Dash
- Created interactive graphs
- User can filter by launch site
- Helps in better understanding data

Machine Learning



- Models Used:
- Logistic Regression
- Decision Tree
- SVM
- Goal:
Predict landing success

Accuracy



- Accuracy Comparison:
- Logistic: 83%
- Decision Tree: 87%
- SVM: 85%
- Best Model: Decision Tree

Conclusion

- Rocket landing success can be predicted
- Data analysis provides useful insights
- Machine learning improves prediction accuracy
- Decision Tree performed best

Future Insights

- Use more real-time data
- Improve model accuracy
- Deploy as web application
- Add more features

THANK YOU